MODULE 4 | AIRCRAFT FLIGHT CONTROLS

ASYS 245 | PART A | ATA 27 | CORNELL NOTES

SECTION 2 • Control-System Actuation

CUES / QUESTIONS

1. What four control-actuation methods are introduced?

2. How are mechanical cables routed to the flight controls?

3. How do push-pull rods transmit motion, and what must be inspected?

4. What is hydraulic assistance and what happens if pressure is lost?

5. How does fly-by-wire transmit commands to control surfaces?

NOTES

1 | FLIGHT-CONTROL ACTUATION: OVERVIEW

The presentation lists four approaches for connecting the flight deck to aerodynamic control surfaces:

CABLE SYSTEM | PUSH-PULL RODS | HYDRAULIC ASSIST | FLY-BY-WIRE

2 | CABLE OPERATED SYSTEM

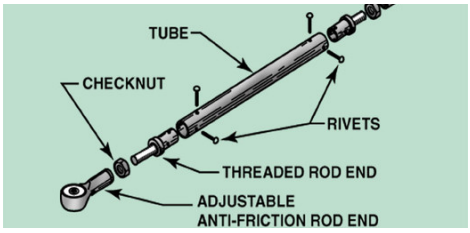


Steel cables link cockpit controls to the flight control surfaces. Cables are routed through pulleys and fairleads.

The cable arrangement provides mechanical control; the source also describes some systems with hydraulic assistance.

PILOT INPUT → STEEL CABLES → PULLEYS → CONTROL SURFACE

3 | PUSH-PULL ROD OPERATED SYSTEM



Aluminum-alloy tubes transmit push and pull movements directly. Adjustable threaded rod ends and jam nuts are used for rigging.

Inspect wear, rod ends, jam nuts, correct engagement depth and the check hole. Wear across many joints can accumulate.

CHECK: rod-end depth • jam nuts • joint condition • rigging • wear

4 | HYDRAULICALLY ASSISTED MECHANICAL CONTROL

Hydraulics assist a mechanical linkage. Mechanical operation can remain possible if hydraulic power is lost.

PILOT → MECHANICAL LINK → HYDRAULIC HELP → SURFACE

5 | FLY-BY-WIRE

Flight-deck inputs become electrical signals that pass through flight-control computers to actuators, typically hydraulic. There is no direct mechanical connection from the flight deck.

INPUT SENSORS → COMPUTER → ACTUATORS → SURFACE

SECTION 2 | STUDENT SUMMARY

- Cable systems use strong steel cables, pulleys and fairleads.
- Hydraulic assistance reduces effort; mechanical linkage remains.
- Push-pull rods require rigging checks and inspection for wear.
- Fly-by-wire sends signals through computers to actuators.